

ECOFLEET AI: INTELLIGENT ELECTRIC VEHICLE FLEET MONITORING AND ROUTE OPTIMIZATION PLATFORM

CSA1026 SOFTWARE ENGINEERING

ARCHITECTURE DIAGRAM

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1.INTRODUCTION

1.1 Introduction

The rapid advancement of electric vehicle (EV) technology has transformed the transportation sector by promoting environmentally sustainable mobility solutions. Governments and industries across the world are encouraging the adoption of electric vehicles to reduce carbon emissions, minimize dependence on fossil fuels, and improve energy efficiency. As organizations increasingly deploy electric vehicles for logistics, delivery services, and public transportation, managing an entire fleet has become a complex task that requires intelligent monitoring and decision-making capabilities.

Unlike conventional vehicles, electric vehicles depend heavily on battery performance, charging infrastructure, and efficient energy management. Fleet operators must continuously monitor battery health, charging schedules, vehicle availability, energy consumption, and travel routes to ensure uninterrupted operations. Manual monitoring of these activities is time-consuming and often leads to inefficient resource utilization, increased operational costs, and unexpected vehicle downtime.

Recent developments in Artificial Intelligence (AI), Internet of Things (IoT), Global Positioning System (GPS), Cloud Computing, and Machine Learning (ML) have enabled organizations to build intelligent fleet management systems capable of analyzing real-time operational data. These technologies make it possible to monitor vehicle health, predict battery degradation, optimize charging schedules, recommend energy-efficient routes, and generate analytical reports for informed decision-making.

The proposed project, EcoFleet AI: Intelligent Electric Vehicle Fleet Monitoring and Route Optimization Platform, is designed to address these challenges by integrating IoT sensors, AI algorithms, cloud services, and GPS technologies into a unified platform. The system continuously collects vehicle data, processes it using intelligent analytical models, and provides actionable recommendations through an interactive dashboard. By automating fleet monitoring and optimization processes, the platform improves operational efficiency, enhances vehicle utilization, reduces energy consumption, and supports sustainable transportation initiatives.

1.2 Background of the Study

The transportation industry has experienced significant changes with the widespread adoption of electric vehicles in commercial operations. Logistics companies, ride-sharing services, public transport agencies, and delivery organizations are replacing conventional vehicles with electric vehicles to achieve environmental sustainability and comply with government regulations regarding carbon emissions.

Although electric vehicles provide numerous environmental and economic benefits, they introduce new operational challenges. Fleet managers must monitor battery charging cycles, estimate remaining driving range, schedule maintenance, identify nearby charging stations, and optimize routes based on traffic conditions. Existing fleet management systems mainly provide location tracking and vehicle monitoring but often lack intelligent capabilities such as predictive maintenance, battery life estimation, and charging optimization.

Artificial Intelligence and Machine Learning technologies have emerged as powerful tools for improving fleet operations by analyzing historical and real-time vehicle data. These technologies enable predictive analytics, route optimization, and preventive maintenance, allowing organizations to improve fleet productivity while reducing operating costs. Consequently, the integration of AI into EV fleet management has become an essential requirement for developing smart transportation systems.

1.3 Problem Statement

Electric vehicle fleet operators encounter numerous challenges in managing battery performance, charging infrastructure, route planning, and vehicle maintenance simultaneously. Conventional fleet management systems primarily focus on vehicle tracking but lack intelligent capabilities for predicting battery degradation, optimizing charging schedules, and recommending energy-efficient travel routes. As a result, fleet operators may experience unnecessary charging stops, increased operational expenses, reduced battery lifespan, delivery delays, and lower fleet utilization.

The absence of predictive maintenance mechanisms also increases the possibility of unexpected vehicle failures, leading to operational downtime and reduced customer satisfaction. Furthermore, limited visibility into charging station

availability and vehicle health makes it difficult for fleet managers to make timely operational decisions.

Therefore, there is a need for an intelligent fleet monitoring platform capable of continuously collecting real-time vehicle data, predicting maintenance requirements, optimizing charging operations, and recommending efficient travel routes using Artificial Intelligence and Machine Learning techniques. Such a system can significantly improve fleet productivity, reduce energy consumption, minimize operational costs, and contribute toward sustainable transportation.

1.4 Objectives

The primary objective of the proposed EcoFleet AI platform is to design and develop an intelligent software system that enhances the efficiency, reliability, and sustainability of electric vehicle fleet operations through real-time monitoring and AI-based decision support.

The specific objectives of the proposed system are:

- To develop an AI-powered platform for monitoring electric vehicle fleet performance.
- To continuously monitor battery health, energy consumption, and vehicle status using IoT sensors.
- To predict battery degradation and maintenance requirements using Machine Learning algorithms.
- To optimize charging schedules by considering battery condition, electricity demand, and charging station availability.
- To recommend energy-efficient routes using GPS data, traffic information, and vehicle performance analysis.
- To provide interactive dashboards for fleet managers to monitor vehicle operations in real time.
- To generate analytical reports on fleet utilization, battery efficiency, charging performance, and operational costs.
- To improve fleet productivity while reducing operational expenses, energy consumption, and environmental impact.

1.5 Scope of the Project

The scope of the EcoFleet AI platform covers the design and development of an intelligent software solution for monitoring and optimizing electric vehicle fleets operating in logistics companies, public transportation systems, corporate transportation services, and commercial delivery organizations.

The system integrates IoT sensors installed in electric vehicles to continuously collect operational information, including battery charge level, battery health, vehicle location, energy consumption, charging status, and travel history. These data are transmitted to a cloud-based platform where Artificial Intelligence and Machine Learning algorithms analyze the information to predict battery degradation, recommend preventive maintenance, optimize charging schedules, and determine the most energy-efficient routes.

The platform also includes a centralized dashboard that allows fleet managers to monitor vehicle status, charging station availability, maintenance alerts, route recommendations, and fleet performance through graphical reports. The architecture is designed to support scalability, maintainability, reliability, and future integration with autonomous vehicles, renewable energy sources, smart charging infrastructure, and smart city transportation systems.

1.6 Expected Outcomes

The successful implementation of the EcoFleet AI platform is expected to improve both operational efficiency and environmental sustainability. By continuously monitoring battery performance and applying predictive maintenance techniques, the system can extend battery lifespan and reduce unexpected vehicle failures.

Intelligent charging optimization will minimize electricity costs while ensuring maximum vehicle availability. AI-based route optimization will reduce travel distance, avoid unnecessary charging stops, and improve delivery efficiency. Fleet managers will gain better operational visibility through real-time dashboards and comprehensive analytical reports, enabling faster and more informed decision-making.

Furthermore, the platform will contribute to reducing carbon emissions by promoting efficient electric vehicle operations and supporting the adoption of sustainable transportation technologies.

2.SYSTEM REQUIREMENT ANALYSIS

2.1 Introduction

System Requirement Analysis is the process of identifying and documenting the requirements of a software system before design and development. It helps ensure that the proposed system meets stakeholder expectations and business objectives. For the EcoFleet AI platform, the analysis focuses on monitoring electric vehicles, optimizing charging schedules, predicting battery health, and improving fleet efficiency through AI and IoT technologies.

2.2 Stakeholder Analysis

The EcoFleet AI platform involves multiple stakeholders, each performing different responsibilities.

Stakeholder	Responsibilities
Fleet Manager	Monitor vehicles, battery health, routes, and reports.
Administrator	Manage users, vehicles, and system settings.
Driver	View optimized routes and receive alerts.
Maintenance Engineer	Monitor battery health and schedule maintenance.
GPS&Charging Services	Provide location, traffic, and charging station information.

2.3 Functional Requirements

The following table summarizes the major functional requirements of the proposed system.

ID	Functional Requirement
FR-01	User Login and Authentication
FR-02	Vehicle Registration
FR-03	Real-Time Fleet Monitoring
FR-04	Battery Health Monitoring

FR-05	GPS Tracking
FR-06	Route Optimization
FR-07	Charging Schedule Management
FR-08	Predictive Maintenance
FR-09	Dashboard and Reports
FR-10	Notification Alerts

2.4 Non-Functional Requirements

Requirement	Description
Performance	Process real-time vehicle data efficiently.
Scalability	Support a growing number of vehicles and users.
Security	Protect data using authentication and encryption.
Reliability	Ensure accurate and continuous monitoring.
Availability	Provide 24/7 system access.
Maintainability	Support easy updates and future enhancements.

2.5 Requirement Prioritization

The requirements are prioritized based on their importance to the system.

Requirement	Priority
Fleet Monitoring	Highest
Battery Health Monitoring	Highest
Route Optimization	Highest
Charging Management	High
Predictive Maintenance	High
Dashboard	Medium

Report Generation	Medium
Notifications	Medium

2.6 Quality Attributes

The proposed system is designed with the following quality attributes:

- Scalability: Supports expansion as the fleet size increases.
- Performance: Processes IoT and GPS data in real time.
- Security: Protects user and vehicle data through secure authentication and encryption.
- Reliability: Ensures continuous monitoring with minimal downtime.
- Availability: Provides uninterrupted access for fleet managers.
- Maintainability: Modular design allows easy updates and feature additions.

2.7 Requirement Traceability Matrix (RTM)

Requirement	Module
User Authentication	Authentication Module
Fleet Monitoring	Fleet Management Module
Battery Monitoring	Battery Management Module
Route Optimization	AI Route Module
Charging Management	Charging Module
Predictive Maintenance	AI Prediction Module
Dashboard & Reports	Analytics Module

3.REQUIREMENT ANALYSIS USING JIRA

3.1 Introduction

Jira is an Agile project management and issue-tracking tool used to organize project requirements, monitor development progress, and manage software tasks efficiently. For the proposed EcoFleet AI: Intelligent Electric Vehicle Fleet Monitoring and Route Optimization Platform, Jira was used to create the project workspace, organize system requirements, prioritize tasks, and monitor project progress using a Kanban Board.

The project requirements were converted into individual work items representing the major functionalities of the system. Each work item was assigned a priority and moved through different workflow stages to simulate the software development process.

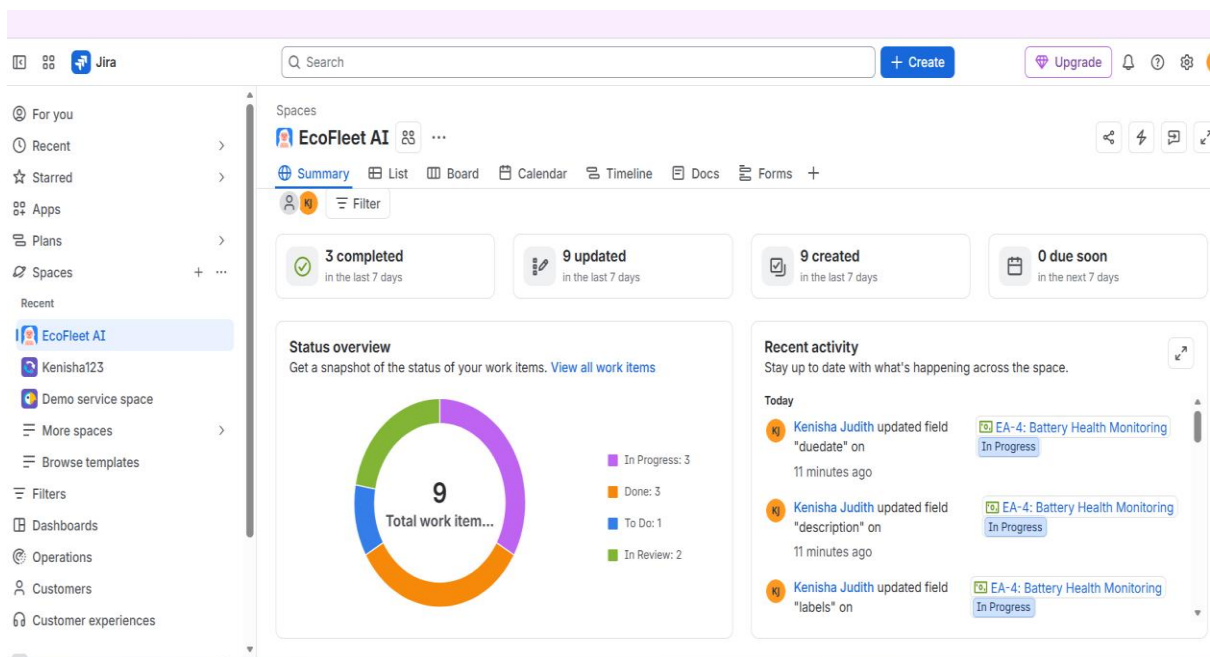


Figure 3.1: Jira Project Summary

3.2 Product Backlog

The Product Backlog contains the list of requirements identified during the requirement analysis phase. Each work item represents an important feature of the EcoFleet AI platform. The backlog was organized according to project requirements and assigned appropriate priorities to support systematic software development.

The major backlog items created in Jira include:

Work Item	Priority
Login Module	Medium
Vehicle Registration	Medium
Fleet Monitoring	High
Battery Health Monitoring	High
Route Optimization	Medium
Charging Schedule	Medium
AI Prediction	Medium
Dashboard	Medium
Report Generation	Medium

The Product Backlog provides a structured view of all functional requirements and helps the development team manage and prioritize project activities effectively.

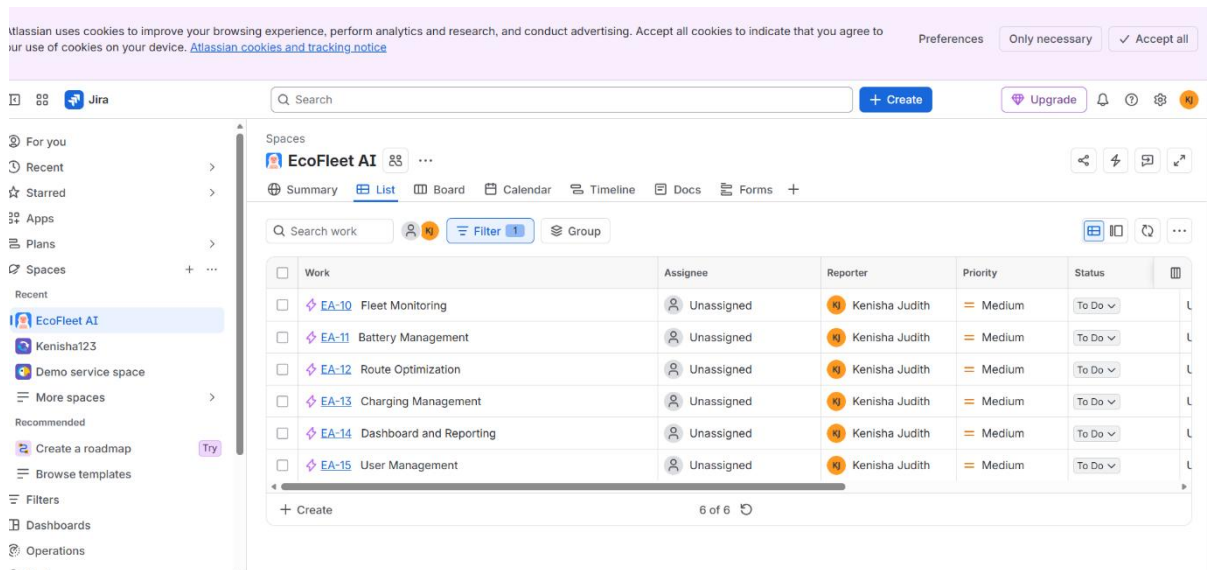


Figure 3.2: Jira Product Backlog

3.3 Kanban Board

The Kanban Board was used to monitor the progress of project tasks throughout the development process. The workflow consists of four stages:

- **To Do** – Tasks that are yet to be started.
- **In Progress** – Tasks currently under development.
- **In Review** – Tasks completed and waiting for verification.
- **Done** – Tasks that have been successfully completed.

The Kanban Board provides a visual representation of project progress and enables efficient tracking of task status. It improves workflow management, enhances collaboration, and ensures that project activities are completed in an organized manner.

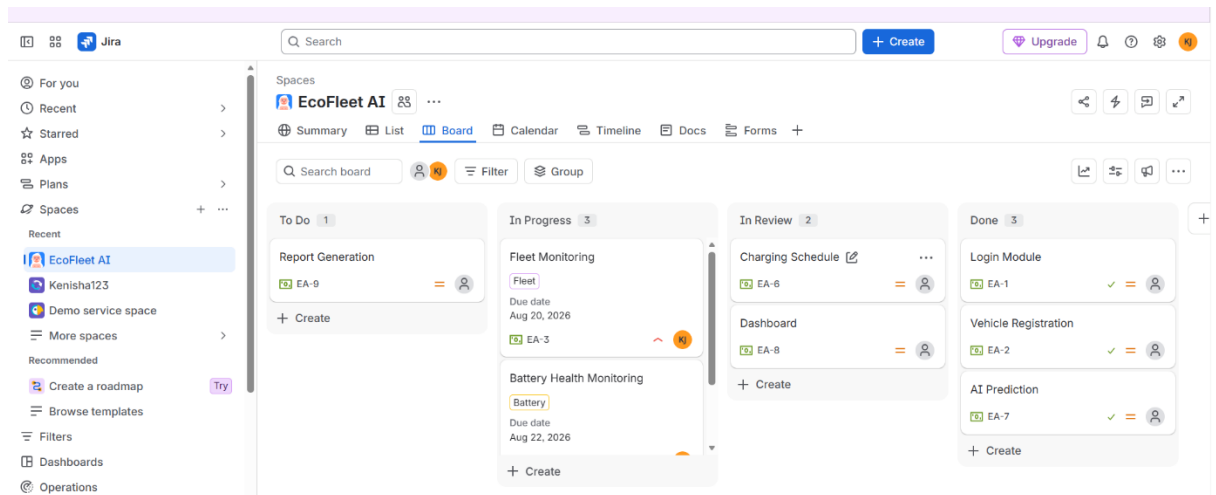


Figure 3.3: Jira Kanban Board

3.4 Benefits of Using Jira

The use of Jira in the EcoFleet AI project offers several advantages:

- Organizes project requirements in a structured manner.
- Helps prioritize software development tasks.
- Enables tracking of task progress using a Kanban Board.
- Improves project planning and workflow management.
- Supports Agile software development practices.
- Enhances collaboration among team members.

3.5 Chapter Summary

Jira was successfully used to manage the requirements of the EcoFleet AI project. The Product Backlog and Kanban Board provided an organized approach for planning, prioritizing, and tracking project activities. The use of Jira supports effective requirement management and serves as a foundation for the software architecture design presented in the following chapter.

4.SOFTWARE ARCHITECTURE SELECTION

4.1 Introduction

Software architecture defines the overall structure of a software system and describes how its components interact to achieve the desired functionality. Selecting an appropriate architecture is essential because it directly influences the system's scalability, performance, security, maintainability, and reliability.

For the proposed **EcoFleet AI: Intelligent Electric Vehicle Fleet Monitoring and Route Optimization Platform**, the architecture must support real-time data collection from IoT devices, Artificial Intelligence (AI) for predictive analytics, GPS-based route optimization, cloud-based data storage, and interactive dashboards. Therefore, a suitable architecture should provide modularity, flexibility, and efficient communication between system components.

4.2 Evaluation of Architectural Styles

Several software architectures were evaluated before selecting the most suitable one for the EcoFleet AI platform.

Architecture	Advantages	Limitations
Layered Architecture	Easy to develop and maintain, clear separation of responsibilities.	Less suitable for real-time IoT communication.
Client–Server Architecture	Centralized management and secure communication between clients and servers.	Difficult to scale for multiple independent services.
Microservices Architecture	Independent services, high scalability, easy deployment and maintenance.	More complex to manage and deploy.
Hybrid Architecture	Combines the strengths of Layered, Client–Server, and Microservices architectures.	Requires careful integration between services.

4.3 Selected Architecture

After analyzing the system requirements, **Hybrid Architecture** was selected for the EcoFleet AI platform.

The Hybrid Architecture combines the advantages of **Layered Architecture**, **Client–Server Architecture**, and **Microservices Architecture**. The layered structure provides a clear separation between the user interface, business logic, and data management. The client–server model enables communication between users and the cloud platform, while microservices allow individual services such as battery monitoring, route optimization, charging management, and AI prediction to operate independently.

This combination makes the platform more scalable, maintainable, and suitable for handling real-time IoT data and AI-based processing.

4.4 Justification for Selecting Hybrid Architecture

The Hybrid Architecture was selected because it satisfies the functional and non-functional requirements identified during the requirement analysis. The following reasons justify its selection:

- Supports real-time communication between IoT devices and the cloud platform.
- Enables independent services for battery monitoring, route optimization, charging management, and predictive maintenance.
- Improves scalability by allowing new services to be added without affecting existing components.
- Enhances maintainability through modular system design.
- Provides better reliability by isolating failures within individual services.
- Facilitates integration with external systems such as GPS services and charging station APIs.
- Supports future enhancements such as autonomous fleet management and renewable energy integration.

4.5 Benefits of the Selected Architecture

The Hybrid Architecture provides several benefits for the EcoFleet AI platform:

- **Scalability:** New vehicles and services can be added without significant system modifications.
- **Performance:** Independent services process real-time data efficiently.
- **Security:** Authentication and secure communication protect sensitive fleet information.
- **Maintainability:** Individual modules can be updated or replaced independently.
- **Reliability:** Failure in one module does not affect the entire system.
- **Flexibility:** The architecture supports integration with future technologies and external APIs.

4.6 Chapter Summary

This chapter evaluated different software architectural styles and selected the **Hybrid Architecture** as the most suitable solution for the EcoFleet AI platform. The selected architecture combines the strengths of Layered, Client–Server, and Microservices architectures, providing a scalable, secure, and maintainable foundation for real-time fleet monitoring, AI-based prediction, and route optimization. The detailed architecture design and component interactions are presented in the next chapter.

5.SOFTWARE ARCHITECTURE DESIGN

5.1 Introduction

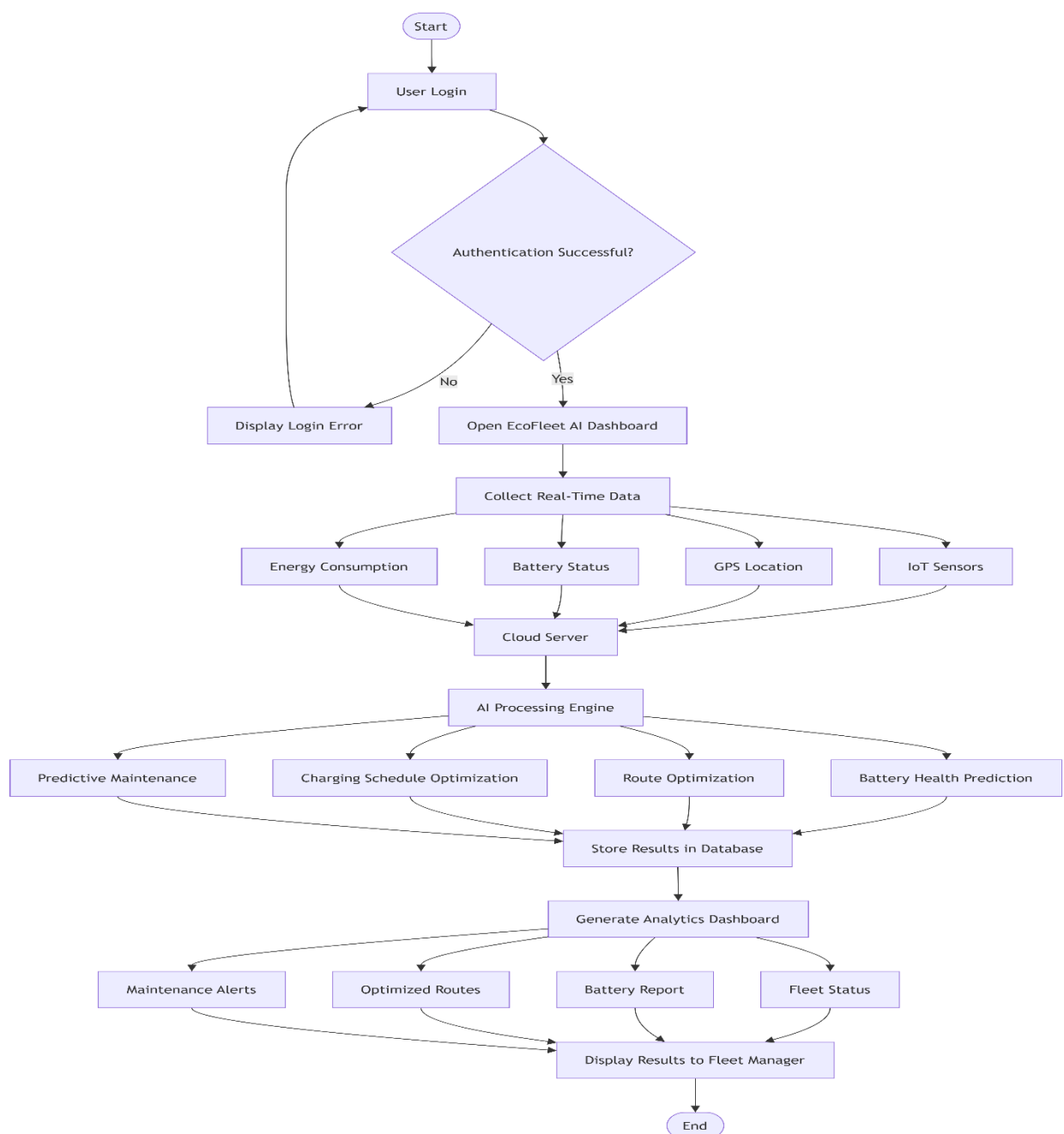
The software architecture design defines the overall structure of the EcoFleet AI platform and illustrates how different components communicate to provide intelligent fleet management services. The architecture integrates IoT devices, Artificial Intelligence (AI), GPS services, cloud computing, and databases to support real-time monitoring, predictive maintenance, charging optimization, and route planning.

The proposed system follows a **Hybrid Architecture**, combining layered and microservice principles. This approach improves scalability, maintainability, and flexibility while allowing independent modules to process real-time vehicle data efficiently.

5.2 Software Architecture Diagram

The architecture consists of five major layers:

1. Presentation Layer
2. Application Layer
3. AI & Analytics Layer
4. Data Layer
5. External Services Layer



5.3 Description of Architecture Layers

5.3.1 Presentation Layer

The Presentation Layer provides the user interface for administrators, fleet managers, maintenance engineers, and drivers. Users access the platform through a web application or mobile application to monitor fleet operations, view dashboards, receive alerts, and generate reports.

Components:

- Web Dashboard
- Mobile Application
- User Login

5.3.2 Application Layer

The Application Layer contains the core business logic of the system. It processes user requests and coordinates communication between the user interface, AI services, and the database.

Modules:

- User Authentication
- Fleet Monitoring
- Battery Management
- Charging Management
- Route Optimization
- Notification Service
- Report Generation

5.3.3 AI & Analytics Layer

This layer performs intelligent analysis using Artificial Intelligence and Machine Learning algorithms. It predicts battery degradation, recommends charging schedules, analyzes fleet performance, and suggests energy-efficient routes.

Functions:

- Battery Health Prediction
- Predictive Maintenance
- AI Route Optimization
- Energy Consumption Analysis

5.3.4 Data Layer

The Data Layer stores all operational information required by the platform.

It maintains:

- Vehicle details
- User information
- Battery health records
- GPS location history
- Charging records
- Fleet reports

A cloud database ensures secure storage and easy access to real-time data.

5.3.5 External Services Layer

The platform communicates with several external services to obtain real-time information.

These include:

- GPS Navigation Service
- Traffic Information API
- IoT Sensors
- Charging Station API
- Cloud Services

These services provide data required for monitoring, route optimization, and charging management.

5.4 Architecture Workflow

The overall workflow of the EcoFleet AI platform is as follows:

1. IoT sensors installed in electric vehicles continuously collect battery health, vehicle status, speed, and energy consumption data.
2. The collected data are transmitted to the cloud server for processing.
3. The AI module analyzes the data to predict battery degradation, recommend charging schedules, and optimize travel routes.
4. The processed information is stored in the cloud database.
5. Fleet managers access the web dashboard to monitor vehicle performance, receive alerts, and generate analytical reports.
6. Drivers receive optimized routes and charging recommendations through the mobile application.

This workflow enables real-time monitoring, intelligent decision-making, and improved fleet efficiency.

5.5 Chapter Summary

This chapter presented the software architecture design of the EcoFleet AI platform. The Hybrid Architecture integrates user interfaces, application services, AI analytics, cloud storage, and external services into a single intelligent system. The modular design improves scalability, reliability, and maintainability while supporting real-time fleet monitoring and energy-efficient operations.

6.COMPONENT DESCRIPTION AND INTERACTION

6.1 Introduction

The EcoFleet AI platform consists of several software components that work together to provide intelligent electric vehicle fleet monitoring and route optimization. Each component performs a specific function while communicating with other components to achieve efficient fleet management. The modular design improves scalability, maintainability, and system performance.

6.2 User Authentication Module

Purpose

The User Authentication Module ensures secure access to the EcoFleet AI platform. It verifies user credentials and allows only authorized users to access the system.

Inputs

- Username
- Password

Processing

- Validates user credentials.
- Authenticates users.
- Assigns user roles and permissions.

Outputs

- Successful login.
- Access to the system dashboard.
- Error message for invalid credentials.

6.3 Fleet Monitoring Module

Purpose

This module continuously monitors the operational status of electric vehicles in the fleet. It collects real-time information from IoT sensors and GPS devices to track vehicle location, speed, and operational status.

Inputs

- GPS coordinates
- Vehicle status
- Sensor readings

Processing

- Tracks vehicle movement.
- Updates fleet status.
- Identifies inactive vehicles.

Outputs

- Live vehicle status.
- Fleet monitoring dashboard.
- Vehicle location updates.

6.4 Battery Management Module

Purpose

The Battery Management Module monitors battery charge level, health status, charging cycles, and temperature to improve battery life and ensure reliable vehicle operation.

Inputs

- Battery percentage
- Battery temperature
- Charging status

Processing

- Calculates battery health.
- Detects abnormal battery conditions.
- Tracks charging cycles.

Outputs

- Battery health report.
- Battery status updates.
- Low battery alerts.

6.5 AI Prediction Module

Purpose

The AI Prediction Module uses machine learning algorithms to analyze historical and real-time data for predicting battery degradation and maintenance requirements.

Inputs

- Battery history
- Vehicle performance data
- Charging history

Processing

- Predicts battery degradation.
- Detects maintenance requirements.
- Analyzes energy consumption.

Outputs

- Predictive maintenance alerts.
- Battery life estimation.
- Performance analysis.

6.6 Route Optimization Module

Purpose

This module recommends energy-efficient travel routes by considering GPS location, traffic conditions, battery level, and charging station availability.

Inputs

- GPS location
- Traffic information
- Battery level

Processing

- Calculates multiple routes.
- Selects the most energy-efficient route.
- Estimates travel time.

Outputs

- Optimized route.
- Estimated arrival time.

- Route recommendations.

6.7 Charging Management Module

Purpose

The Charging Management Module schedules charging operations based on battery condition, charging station availability, and electricity demand.

Inputs

- Battery level
- Charging station information
- Vehicle schedule

Processing

- Identifies nearby charging stations.
- Schedules charging time.
- Optimizes charging operations.

Outputs

- Charging recommendations.
- Charging schedule.
- Charging station availability.

6.8 Dashboard and Reporting Module

Purpose

This module provides fleet managers with real-time dashboards, graphical reports, and operational analytics for effective decision-making.

Inputs

- Fleet data
- Battery reports
- AI predictions

Processing

- Generates analytical reports.

- Displays performance metrics.
- Summarizes fleet operations.

Outputs

- Fleet utilization report.
- Battery health report.
- Energy consumption report.
- Dashboard analytics.

6.9 Component Interaction

The components of the EcoFleet AI platform work together to ensure efficient fleet management. The User Authentication Module verifies user credentials before granting access to the dashboard. The Fleet Monitoring Module continuously collects real-time vehicle information from IoT sensors and GPS devices. The Battery Management Module evaluates battery condition and forwards the data to the AI Prediction Module for analysis. Based on the analysis, the Route Optimization Module recommends energy-efficient routes, while the Charging Management Module suggests suitable charging schedules and nearby charging stations. Finally, all processed information is stored in the cloud database and displayed through the Dashboard and Reporting Module, enabling fleet managers to monitor vehicle performance and make informed decisions.

6.10 Chapter Summary

This chapter described the major software components of the EcoFleet AI platform and explained their roles and interactions. Each component contributes to intelligent fleet monitoring, battery management, route optimization, charging management, and predictive maintenance. Together, these components provide a scalable, reliable, and efficient solution for electric vehicle fleet management.

7.QUALITY ATTRIBUTE ANALYSIS

7.1 Introduction

Quality attributes define how well a software system performs beyond its functional requirements. For the EcoFleet AI platform, quality attributes ensure efficient fleet monitoring, secure data management, and reliable system performance.

7.2 Quality Attributes

Quality Attribute	Description
Performance	Processes real-time vehicle and battery data efficiently.
Scalability	Supports an increasing number of electric vehicles and users.
Security	Protects user credentials and fleet information through secure authentication.
Reliability	Ensures continuous monitoring with minimal system failures.
Availability	Provides 24/7 access for fleet managers and drivers.
Maintainability	Allows easy updates and addition of new features.
Usability	Offers a simple and user-friendly interface for monitoring fleet operations.

7.3 Chapter Summary

The selected Hybrid Architecture satisfies the required quality attributes by providing a scalable, secure, reliable, and maintainable platform for intelligent electric vehicle fleet management.

8.ARCHITECTURAL DECISIONS AND TRADE-OFFS

8.1 Architectural Decisions

The following design decisions were made while developing the EcoFleet AI platform:

- Hybrid Architecture was selected to combine scalability and modularity.
- AI was integrated to predict battery degradation and maintenance requirements.
- IoT sensors were used for real-time vehicle and battery monitoring.
- GPS services were incorporated for route optimization.
- Cloud databases were selected for centralized data storage.

8.2 Trade-Off Analysis

Decision	Benefit	Limitation
Hybrid Architecture	Flexible and scalable	More complex than a simple layered architecture
AI Prediction	Improves maintenance planning	Requires historical data
IoT Integration	Provides real-time monitoring	Additional hardware cost
Cloud Database	Centralized and scalable storage	Depends on internet connectivity

8.3 Chapter Summary

The architectural decisions improve system performance, scalability, and maintainability while introducing manageable trade-offs that are acceptable for the proposed solution.

9.FUTURE ENHANCEMENTS

The EcoFleet AI platform can be further improved through the following enhancements:

- Develop Android and iOS mobile applications.
- Integrate renewable energy-based charging stations.
- Introduce voice-enabled fleet management.
- Include weather-aware route optimization.
- Apply blockchain technology for secure charging transactions.
- Enhance AI models for more accurate battery life prediction.

10.CONCLUSION

The EcoFleet AI platform provides an intelligent solution for monitoring electric vehicle fleets, predicting battery health, optimizing charging schedules, and recommending energy-efficient routes. The proposed Hybrid Architecture ensures scalability, security, and maintainability while supporting real-time fleet operations. The use of AI, IoT, GPS, and cloud technologies enables efficient fleet management and contributes to sustainable transportation. The proposed system can be further enhanced through future technological advancements to support larger fleets and smarter mobility solutions.